

Declan Chan

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Summary

Engineering Science student at the University of Toronto with experience in machine learning research, aerospace systems, CAD design, and software development.

Skills and Proficiencies

Programming: Python, C, MATLAB, HTML/CSS, Git

Design: SolidWorks, OnShape

Education

Engineering Science – University of Toronto

2025–Present

- 2× Dean's Honours List

International Baccalaureate Diploma Programme

2021–2025

- IB Score: 45/45
- OSSD Top 6: 99.67%
- Director's Achievement Award

Experiences

Machine Learning Researcher – AutoDIAL

2026–Present

- Autonomous Discovery of Alloys (AutoDIAL) researcher.
- Performed adversarial attacks on Machine Learning Force Field models to test their robustness.
- Benchmarked the failure points of Message Passing Atomic Cluster Expansion training models.
- Implemented testing workflows using Python, PyTorch, and Jupyter Notebook.

Space Systems Engineer – University of Toronto Aerospace Team

2026–Present

- Contributed to CubeSat attitude determination and control systems.
- Simulated magnetic environments using a Helmholtz cage.

Senior Karate Instructor – Northern Karate Schools

2021–Present

- Yondan: Certified 4th Degree Black Belt.
- Certification: Blended Standard First Aid & CPR/AED Level C.
- Gold medal at the Toronto International Tournament of Martial Arts Champions.
- Teach 100+ kids, teens, and adults every week alongside private classes.

Mechanical Design Engineer – Robotics for Space Exploration

2025–2026

- Designed robotic arm components using CAD for 3D modelling.
- Performed Finite Element Analysis using SolidWorks Simulation to evaluate designs.

NFT Automation

2021–2022

- Programmed a macro to create NFTs on OpenSea.io with Ethereum.
- Explored the Polygon Ethereum blockchain scaling solution.

Technical Projects

GenAI Genesis Hackathon

2026

- Developed an interactive platform visualizing human disease progression as a team of four.
- Implemented a RAG system trained on PubMed data via IBM watsonx.ai.
- Leveraged Agentic AI with React and three.js to build a 3D web application.

Civil Engineering Bridge Competition

2025

- Optimized structural performance and prototyped a matboard bridge using Python.
- Placed 2nd out of 75 teams in Engineering Science as a team of four.

Yarkovsky Effect Orbital Analysis

2024–2025

- Explored the existence of the Yarkovsky effect from asteroids' physical and orbital properties.
- Conducted research and physics calculations under teacher mentorship.

Hydrolysis Investigation ○ Investigated the effect of temperature on the rate constant in the hydrolysis of ethyl acetate. ○ Collected data by conducting experiments using a personalized lab methodology.	2024–2025
Rotational Dynamics Study ○ Studied how the velocity of a volleyball serve affects the ball's angular velocity. ○ Modelling the Magnus effect through calculations and experimental data.	2023–2024
Violin Modelling ○ Analyzed how the density of a violin affects its market price. ○ Used integration to model and calculate volumes of irregular shapes.	2023–2024
Leadership & Activities	
Rubik's Cube Competitions ○ Top 90 in Canada for 3x3x3 fewest moves. ○ Volunteered as a judge for the World Cube Association. ○ Club President who taught beginners and organized school-wide competitions.	2022–2025
Varsity Volleyball ○ Team Captain in Grade 10 and 12. ○ Tier 1 YRAA Championship junior varsity second place. ○ Tier 1 YRAA Championship senior varsity second place.	2022–2025
Rep Volleyball ○ Team Captain and 2× MVP. ○ Silver medal in Canada's 2023 Youth Nationals.	2019–2023
Music Performance ○ Theory and Instrumental: Level 10 RCM comprehensive certificate (First Class Honors). ○ Level 10 RCM practical piano certificate (First Class Honors with Distinction). ○ Level 8 RCM practical violin certificate (First Class Honors with Distinction).	2014–2023